

Android Viva – Short Q&A

1. What is an Activity?

Answer: An Activity represents a single screen in an Android application where the user can interact with the app.

2. What is a Fragment?

Answer: A Fragment is a reusable part of an Activity's UI. It has its own lifecycle and can be used to create modular screens.

3. Why use Fragments?

Answer: Fragments make the application more modular and allow different UI sections to be reused and managed separately.

4. What is RecyclerView?

Answer: RecyclerView is an Android component used to efficiently display a large list of items by reusing item views.

5. Why use RecyclerView instead of a normal ListView?

Answer: RecyclerView provides better performance, flexible layouts, and a ViewHolder mechanism for reusing views.

6. What is an Adapter?

Answer: An Adapter connects the data source with the RecyclerView and creates and binds the item views.

7. What is ViewHolder?

Answer: ViewHolder holds references to the views inside a RecyclerView item so they don't need to be repeatedly searched.

8. What is onCreateViewHolder () ?

Answer: It creates the layout for a new RecyclerView item.

9. What is onBindViewHolder () ?

Answer: It binds the data of a specific item to the views inside the ViewHolder.

10. What is `getItemCount()` ?

Answer: It returns the number of items that the RecyclerView should display.

11. What is an Intent?

Answer: An Intent is used to communicate between Android components, such as opening another Activity.

12. What is `onCreate()` ?

Answer: `onCreate()` is called when an Activity is first created. It is commonly used to initialize views and load initial data.

13. What is `onResume()` ?

Answer: `onResume()` is called when the Activity becomes visible and ready for user interaction.

14. What is `onPause()` ?

Answer: `onPause()` is called when the Activity is no longer in the foreground.

15. What is Retrofit?

Answer: Retrofit is an Android networking library used to communicate with REST APIs.

16. Why is Retrofit used in LankaFresh?

Answer: Retrofit is used to connect the Android application with the backend API and retrieve product and category data.

17. What is an API?

Answer: An API allows different software systems to communicate with each other. In LankaFresh, the Android app communicates with the backend through APIs.

18. What is JSON?

Answer: JSON is a lightweight data format commonly used to exchange data between the Android application and backend.

19. What is Gson?

Answer: Gson converts JSON data into Java objects and Java objects into JSON.

20. What is Glide?

Answer: Glide is an Android image-loading library used to load and display images efficiently.

21. Why use Glide?

Answer: Glide handles image loading, caching, and displaying images efficiently, especially when images come from URLs.

22. What is a Model class?

Answer: A Model class represents the structure of data used by the application, such as `ProductModel` and `CategoryModel`.

23. What is @SerializedName?

Answer: `@SerializedName` tells Gson which JSON field should be mapped to a Java variable.

24. What is RecyclerView.Adapter?

Answer: It is the adapter class responsible for creating item views and connecting model data with the RecyclerView.

25. What is LinearLayoutManager?

Answer: It arranges RecyclerView items in a linear direction, either vertically or horizontally.

26. What is GridLayoutManager?

Answer: It displays RecyclerView items in a grid. In LankaFresh, it can be used to display products in two columns.

27. Why use `GridLayoutManager(this, 2)`?

Answer: The number 2 means the products are displayed in two columns.

28. How does category filtering work?

Answer: When a category is selected, its category ID is compared with the category ID of each product, and only matching products are displayed.

29. Why keep `fullProductList`?

Answer: It keeps the complete product list so products can be filtered without requesting the data from the server again.

30. What happens when a category is clicked?

Answer: The category ID is passed to the filtering method, matching products are selected, and the RecyclerView is updated with the filtered list.

31. What is `notifyDataSetChanged()`?

Answer: It tells the RecyclerView that its data has changed and that the displayed items should be refreshed.

32. What is a Callback in Retrofit?

Answer: A Callback handles the result of an asynchronous API request, including successful responses and failures.

33. What is `onResponse()`?

Answer: `onResponse()` is called when the server sends a response to the API request.

34. What is `onFailure()`?

Answer: `onFailure()` is called when the API request fails because of a network or communication problem.

35. What is `ViewPager2`?

Answer: ViewPager2 allows the user to swipe between pages or slides. In LankaFresh, it can be used for the banner slider.

36. Why use `Handler` in the banner slider?

Answer: A Handler can execute code after a specified delay, allowing the banners to automatically change every few seconds.

37. What is `ArrayList`?

Answer: ArrayList is a dynamic collection used to store multiple objects, such as products or categories.

38. Why use `List<ProductModel>`?

Answer: It stores a collection of product objects and allows the application to process and display multiple products.

39. What is `setAdapter()`?

Answer: `setAdapter()` connects an Adapter to a RecyclerView so that the RecyclerView can display the data.

40. What is `findViewById()`?

Answer: `findViewById()` is used to find a UI component from the Activity layout using its ID.

41. What is `setContentView()`?

Answer: `setContentView()` sets the XML layout that should be displayed by an Activity.

42. What is Android Manifest?

Answer: AndroidManifest.xml contains important application information such as Activities, permissions, services, and other components.

43. Why is Internet permission required?

Answer: Because the LankaFresh Android application communicates with the backend server through the network.

44. What is Context?

Answer: Context provides access to application and Android system resources and is commonly used when creating UI components or accessing resources.

45. Why pass Context to an Adapter?

Answer: The Adapter needs Context for operations such as inflating layouts and loading resources.

Code Explanation Questions

46. Examiner shows:

```
CategoryModel c = list.get(position);
```

Question: What does this do?

Answer: It gets the category object at the current RecyclerView position.

47. Examiner shows:

```
holder.txtName.setText(c.getName());
```

Answer: It displays the category name inside the TextView.

48. Examiner shows:

```
productAdapter.updateList(filteredList);
```

Answer: It sends the filtered products to the ProductAdapter so the RecyclerView can display them.

49. Examiner shows:

```
if (p.getCategoryId() == categoryId)
```

Answer: It checks whether the product belongs to the selected category.

50. Examiner shows:

```
apiService.getProducts().enqueue(...)
```

Answer: It sends an asynchronous API request to retrieve products from the backend.

51. Examiner shows:

```
Glide.with(context)
    .load(fullUrl)
    .into(holder.img);
```

Answer: Glide loads the image from the given URL and displays it in the ImageView.

52. Examiner shows:

```
if (response.isSuccessful() && response.body() != null)
```

Answer: It checks whether the API request was successful and whether the response contains data.

53. Examiner shows:

```
productRecycler.setLayoutManager(
    new GridLayoutManager(this, 2)
);
```

Answer: It configures the product RecyclerView to display two products in each row.

54. Examiner shows:

```
private List<ProductModel> fullProductList =
    new ArrayList<>();
```

Answer: It creates a list to store all products received from the API.

55. Examiner shows:

```
txtStatus.setVisibility(View.GONE);
```

Answer: It hides the status TextView from the screen.

56. Examiner shows:

```
txtStatus.setVisibility(View.VISIBLE);
```

Answer: It makes the status TextView visible.

57. Examiner shows:

```
new Callback<List<ProductModel>>()
```

Answer: It defines a callback for receiving a list of ProductModel objects from the API.

58. Examiner asks: Why do we use ProductModel instead of directly using UI components?

Answer: ProductModel represents the product data separately from the UI, which keeps the code organized and easier to manage.

59. Examiner asks: What is the difference between Product and ProductModel?

Answer: They are separate model classes used for different purposes in the project. ProductModel represents API product data, while Product represents another local product data structure used by the application.

60. Examiner asks: Explain the overall Android data flow.

Answer:

```
Backend API
  ↓
  Retrofit
  ↓
  JSON Response
  ↓
  ProductModel / CategoryModel
  ↓
  Adapter
  ↓
  RecyclerView
  ↓
  Android UI
```

Short explanation: The Android app requests data from the backend using Retrofit. The JSON response is converted into model objects, and the adapters display those objects inside RecyclerViews.

